Problem 5141

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Since $f(y) = y^2 - a^2$, the equation $|f(g(x))| = k$ becomes

$$|g(x)^2 - a^2| = k.$$

For this to have any solutions at all we need $k \geq 0$. The equation then splits into

$$g(x)^2 = a^2 + k \quad \text{or} \quad g(x)^2 = a^2 - k.$$

Each case further splits into $g(x) = \pm\sqrt{a^2 + k}$ and $g(x) = \pm\sqrt{a^2 - k}$ (when the right-hand side is non-negative).

Since g is odd, for every $c \neq 0$ the map $x_0 \mapsto -x_0$ sends solutions of $g(x) = c$ bijectively to solutions of $g(x) = -c$, so those two equations always have the same number of real solutions. Moreover, a solution of $g(x) = c$ cannot simultaneously satisfy $g(x) = -c$ when $c \neq 0$. Hence every nonzero target level contributes an even number of solutions to $|f(g(x))| = k$. Since the total number of solutions is seven (odd), at least one target level must equal 0. Since $a^2 + k \geq a^2 > 0$, the first case contributes a nonzero target, so it must be the second case that supplies the zero level, forcing

$$a^2 - k = 0, \qquad \text{i.e.} \qquad k = a^2.$$

With $k = a^2$ the two cases reduce to $g(x)^2 = 2a^2$ and $g(x)^2 = 0$, giving the three target levels 0 and $\pm a\sqrt{2}$.

Case $g(x) = 0$. Since $g(x) = x(x + 2b)(x - 2b)$, this gives $x \in \{0, \pm 2b\}$: three solutions.

Case $g(x) = \pm a\sqrt{2}$. We need exactly four more solutions. Since g is odd, the map $x \mapsto -x$ sends solutions of $g(x) = a\sqrt{2}$ bijectively to solutions of $g(x) = -a\sqrt{2}$, so the two equations contribute equally and each must account for exactly two solutions.

To determine when $g(x) = c$ has exactly two solutions, we locate g's critical points. Since $b > 0$, setting $g'(x) = 3x^2 - 4b^2 = 0$ gives $x = \pm 2b/\sqrt{3}$. Because $g''(x) = 6x$, we have $g''(-2b/\sqrt{3}) = -12b/\sqrt{3} < 0$, confirming $x = -2b/\sqrt{3}$ is a local maximum, and $g''(2b/\sqrt{3}) > 0$, confirming $x = 2b/\sqrt{3}$ is a local minimum. Since g is odd, the local minimum value equals $-M$ where

$$\begin{aligned} M = g\left(-\frac{2b}{\sqrt{3}}\right) &= \left(-\frac{2b}{\sqrt{3}}\right)^3 - 4b^2\left(-\frac{2b}{\sqrt{3}}\right) \\ &= -\frac{8b^3}{3\sqrt{3}} + \frac{8b^3}{\sqrt{3}} = \frac{16b^3}{3\sqrt{3}} > 0. \end{aligned}$$

Since g is a cubic with exactly two critical points, each a simple local extremum, a horizontal line $y = c$ meets the graph in three points when $|c| < M$, in exactly two points when $|c| = M$ (tangent at one critical point, transverse at the other), and in one point when $|c| > M$. For $g(x) = a\sqrt{2}$ to have exactly two solutions we therefore need $a\sqrt{2} = M$, i.e.

$$a\sqrt{2} = \frac{16b^3}{3\sqrt{3}}.$$

Distinctness of the seven solutions. The three solutions $\{0, \pm 2b\}$ of $g(x) = 0$ are distinct since $b > 0$. The four solutions of $g(x) = \pm a\sqrt{2}$ are disjoint from $\{0, \pm 2b\}$: indeed $g(0) = 0 \neq \pm a\sqrt{2}$ (as $a > 0$), and $g(2b) = (2b)^3 - 4b^2(2b) = 8b^3 - 8b^3 = 0$, with $g(-2b) = 0$ by oddness, so $g(\pm 2b) = 0 \neq \pm a\sqrt{2}$. Hence all seven solutions are distinct.

Final computation. Solving $a\sqrt{2} = 16b^3/(3\sqrt{3})$ for b^3/a:

$$\frac{b^3}{a} = \frac{3\sqrt{3} \cdot \sqrt{2}}{16} = \frac{3\sqrt{6}}{16}.$$